

FACULTY OF ENGINEERING

B.E. (IT/AI/AI&ML/DS/ECE/ETE/Mech/Civil) I - Semester (AICTE) (Backlog) (New)

Examination, July / August 2025

Subject: Engineering Physics

Time: 3 Hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

1.
 - a) Mention four applications of Lasers.
 - b) What do you mean by displacement current? Explain.
 - c) Write any four properties of superconductors.
 - d) What is top-down method?
 - e) Distinguish between classical and quantum bits.
 - f) Write Maxwell's equations in differential form.
 - g) What is meant by magneto-resistance of a material?
2.
 - a) Explain population inversion with a diagram and discuss Einstein's coefficients.
 - b) List the applications of semiconductor devices
3.
 - a) State and prove Poynting theorem
 - b) Discuss the construction and working principle of quantum light emitting diode.
4.
 - a) Explain (i) hysteresis curve (ii) soft and hard magnetic materials.
 - b) Discuss Type-I and Type-II superconductors.
5.
 - a) Describe the properties of nano-materials in detail.
 - b) Distinguish between bulk and thin films. Write the applications of nanomaterials.
6.
 - a) Describe electron beam evaporation method with a neat diagram.
 - b) Obtain an expression for Schrodinger Time-Independent wave equation.
7.
 - a) Describe production of ultrasonic waves by magnetostriction method and write the applications of ultrasonic waves.
 - b) Describe BCS theory of Super Conductivity.
